

Vision foundation models: can they be applied to astrophysics data?

Problem

Scientists are often tasked with solving problems given limited labeled data under distribution shift. Choosing an appropriate foundation model for initial investigation is a non-trivial task.

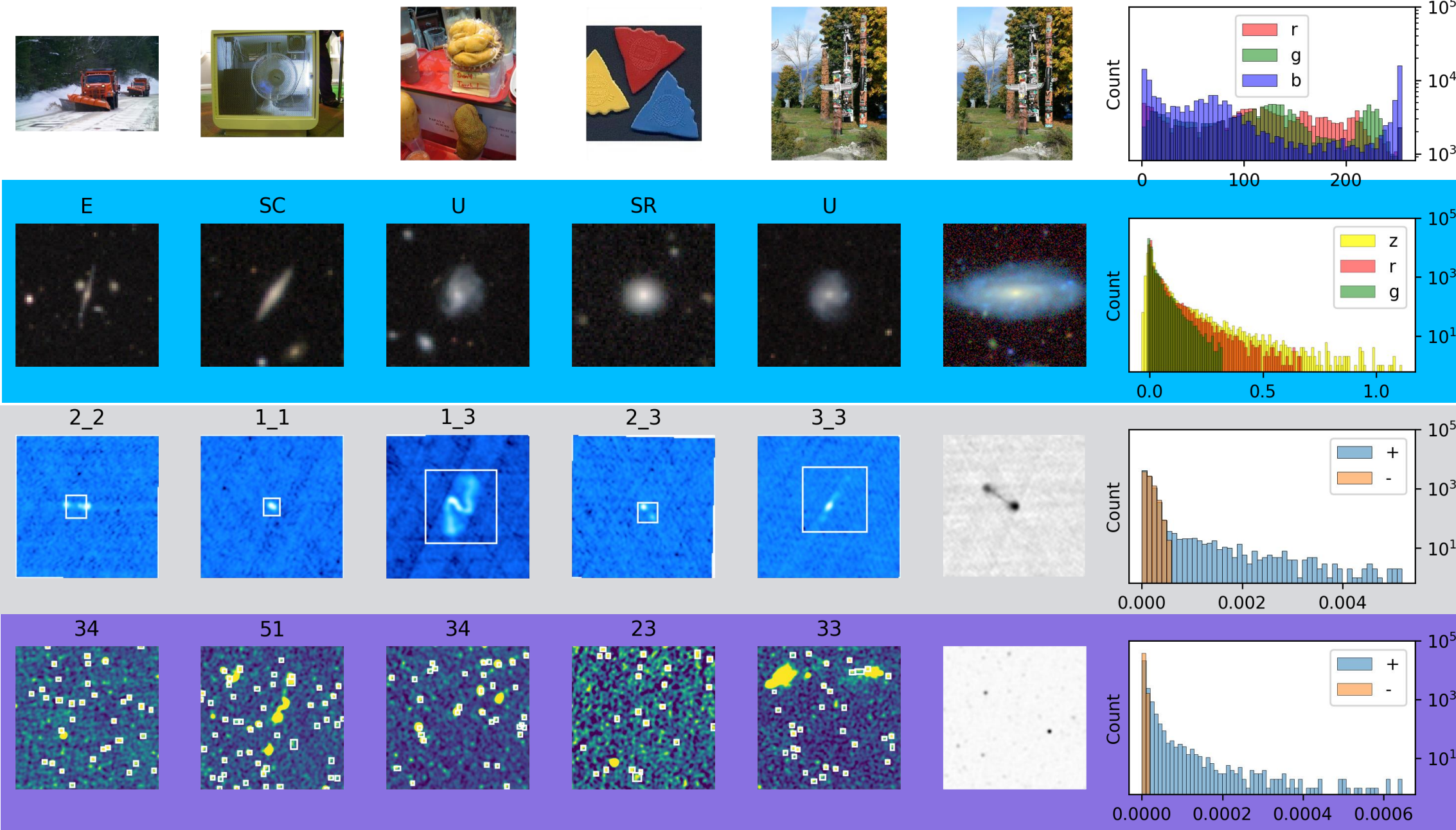
Data

Images from optical and radio astronomy are different from natural images due to:

- Sparseness
- Noise
- High dynamic range
- Artefacts

Therefore, it is commonly assumed that vision foundation models are inadequate for performing tasks in astronomy. Astrophysical images and objectives are neither known nor used during the training of foundation models, which is typically done in a self-supervised way.

We test if the learned features of vision foundation models are suitable for astrophysics images, and how the data, model, and downstream task affect performance.



Dataset examples: ImageNet-1K (top), GalaxyMNIST (second row), Radio Galaxy Zoo (RGZ, third row), MeerKAT MGCLS (bottom). GMNIST and RGZ are labeled according to morphology class, while MGCLS is labeled by the number of compact sources present. White boxes indicate object bounding boxes used for source detection. The rightmost two columns display images in their raw data format, with histograms.

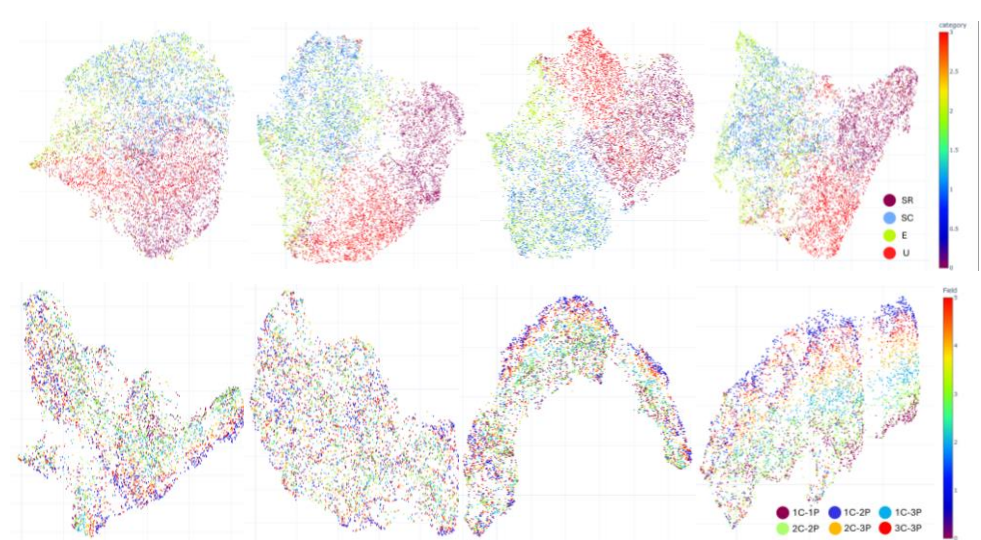
Dataset	Number of Labels	Data Characteristics	Task
GMNIST	10K	centered on bright optical galaxy	classification
RGZ	7.7K	noisy, centered on bright radio galaxy	classification, detection
MGCLS	10K	many radio sources, varying magnitude	source detection

Models

Vision foundation models exhibit significant variability in terms of their architectures, training, and performance characteristics.

Model	Parameters	Pre-training Dataset
MAE*	86M	ImageNet-1k
DINOv2*	86M	LVD-142M
MSN*	86M	ImageNet-1k
ResNet-50	25.6M	ImageNet-1k
ResNet-18	11.5M	ImageNet-1k

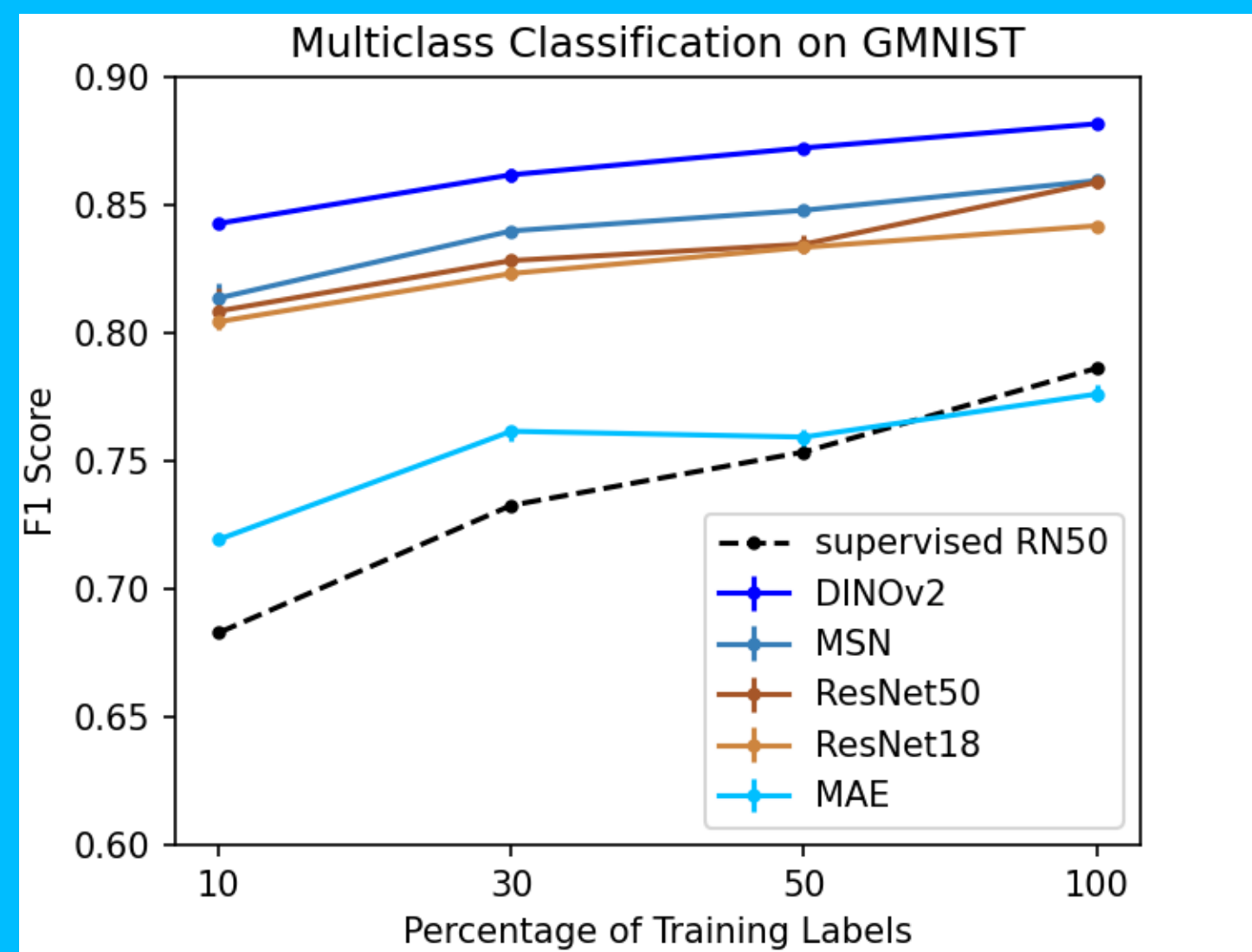
*ViT-Base backbone architecture



UMAP representation of model latent space, for GMNIST (top) and RGZ (bottom) datasets.

Optical Classification Results

Galaxy MNIST (GMNIST) is a balanced dataset of four categories: smooth and round, smooth and cigar-shaped, edge-on-disk, and unbarred spiral. We used models 'out-of-the-box', attaching a single-layer linear classifier head to the pre-trained foundation model backbone. The number of samples used for training was varied at 10%, 30%, 50% and 100%, while the test set was the same. For reference, we also trained a fully supervised ResNet-50.

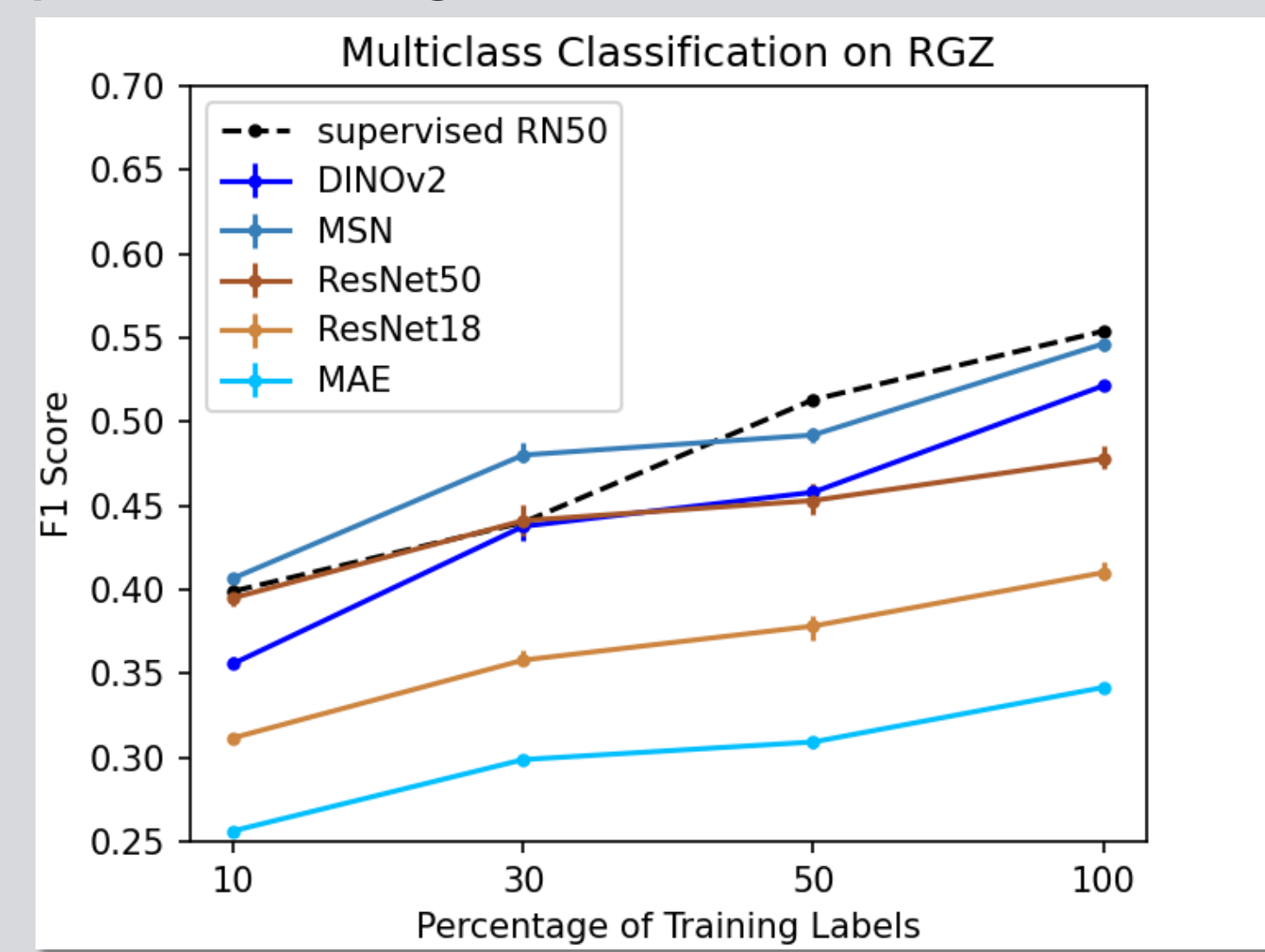


All foundation models out-performed supervised training, improving the classification F1 score up to 15%. Scores of close to 0.85 for 10% labels show that foundation models can be used to great effect for initial investigations for this dataset and task. Of the models tested, DINOv2 and MSN performed the best, along with ResNet-50. MAE, pre-trained using a reconstruction objective, performs worse than the other foundation models.

Shared characteristics of the model pre-training data distribution (all natural images) and GMNIST images might have contributed to good classification performance. GMNIST images are sparser and thus have a distribution skewed towards lower pixel values across all channels. However, like ImageNet, a good portion of central area of the image is occupied by the galaxy of interest.

Radio Classification Results

Radio Galaxy Zoo is unbalanced, with labels determined according to number of distinct radio components (C) and number of intensity peaks (P) in each source. The six possible combinations are: 1C 1P, 1C 2P, 1C 3P, 2C 2P, 2C 3P, and 3C 3P. Classification was difficult, with F1 scores of less than 0.6 even when using all available training labels, **only out-performing or matching supervised training for low numbers of labels**.



RGZ images are Fourier reconstructions containing noise with strong inter-pixel correlations. These statistical differences between natural and radio images suggest that a well-performing model may require knowledge of a very specific set of features; a **true case of distribution shift**.

Foundation models are not immediately applicable to classification on datasets like RGZ; however, the distribution shift can be mitigated. Pre-processing the images so they are closely centered on the galaxy, exploring different fine-tuning techniques, and re-defining the labeling scheme to account for the loss of information about radio components during conversion to PNG all improved classification scores. Both dataset and model constitution, such as ViT patch size relative to the size of the object to be detected or classified, impact performance on downstream tasks.

Full Results

Journal Paper:



Conference Paper:



Code:

